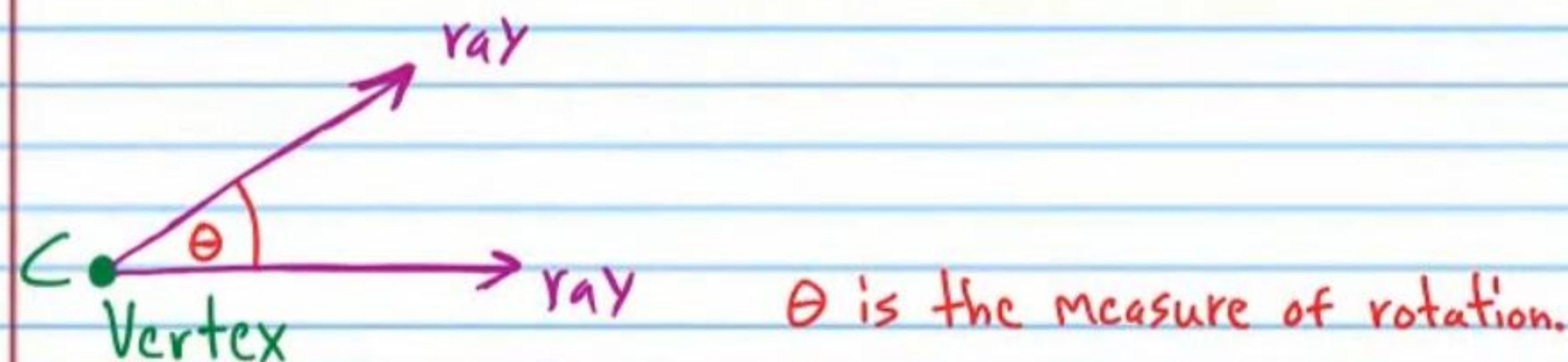


Geometry Review and Trigonometric Expressions with Angle Measures Greater than 90°

Angles

Angle: Two rays with a common endpoint (called the vertex) that help to indicate a measure of rotation.



Angle Vertex: The common endpoint of the two rays.

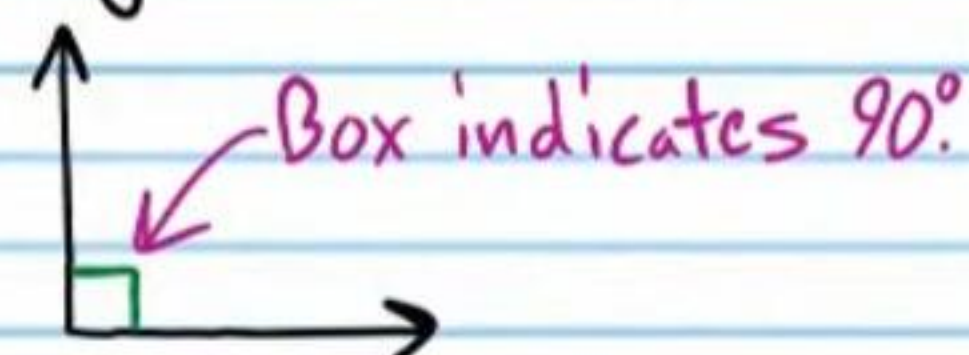
Angle Sides: The two rays.

The measure of an angle is the extent of rotation, and its most common unit is the degree (notated with a "°" superscript). 360° degrees (i.e. 360°) is defined as the measure of a complete rotation.

Classifying Angles

Right Angle

An angle whose measure is 90°



Straight Angle

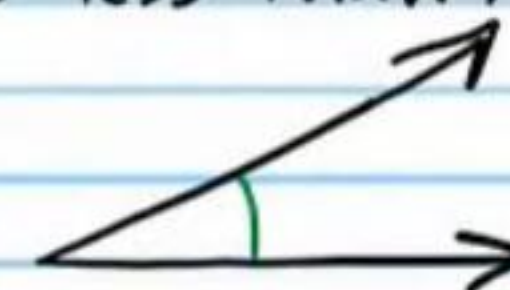
An angle whose measure is 180°

Semicircle indicates 180°



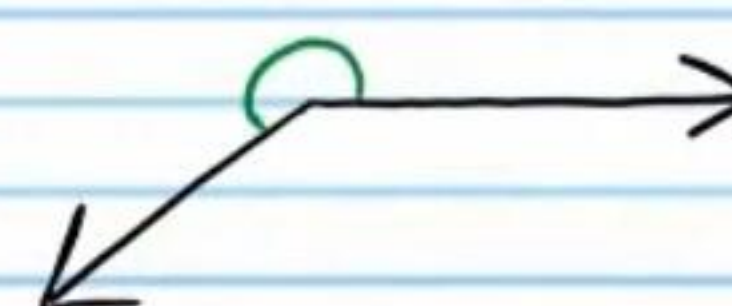
Acute Angle

An angle whose measure is less than 90°



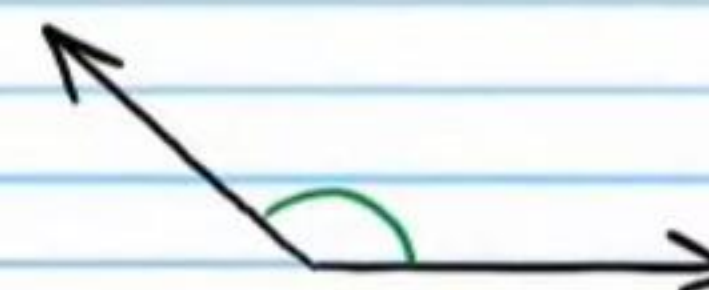
Reflex Angle

An angle whose measure is between 180° and 360°



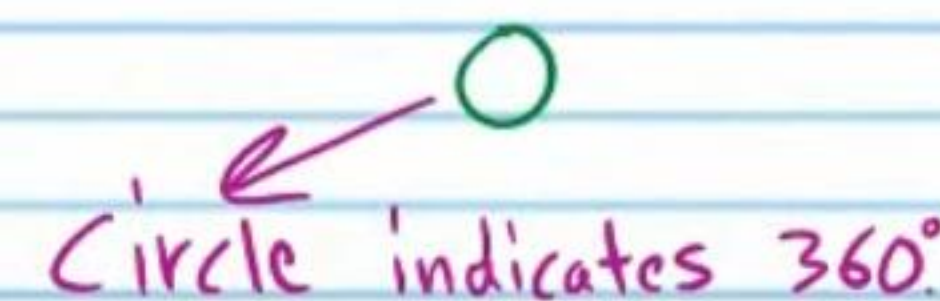
Obtuse Angle

An angle whose measure is between 90° and 180°



Complete Rotation Angle

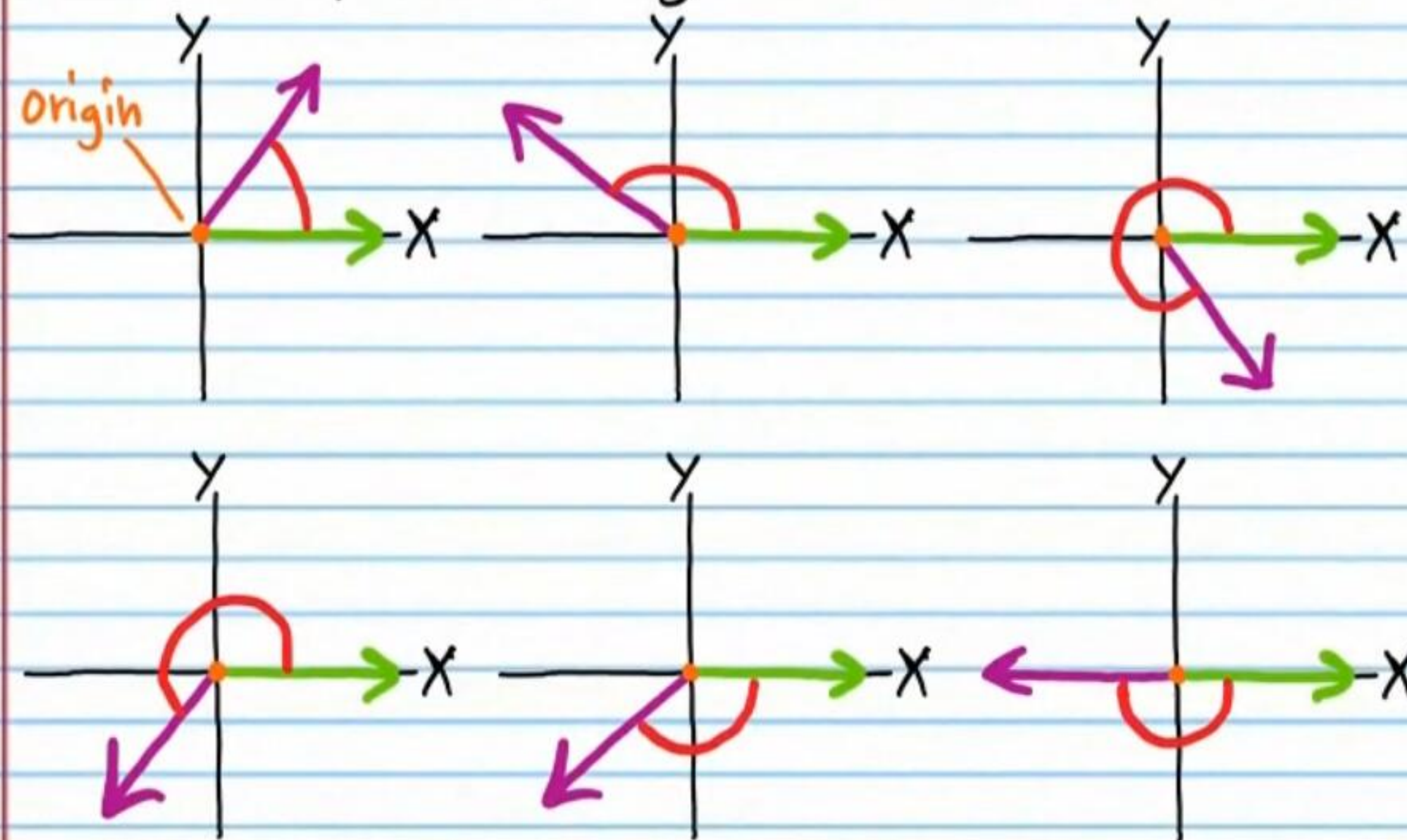
An angle whose measure is 360°



Angles in Standard Position

Standard Position of an Angle: Where an angle is placed so that its vertex is at the origin (i.e. $(0,0)$) of an xy coordinate system and its initial side is on the positive part of the x -axis.

Examples of Angles in Standard Position



Initial Side: The side of the angle that is placed on the positive part of the x -axis.

Terminal Side: The side of the angle where the angle measure stops (i.e. terminates).

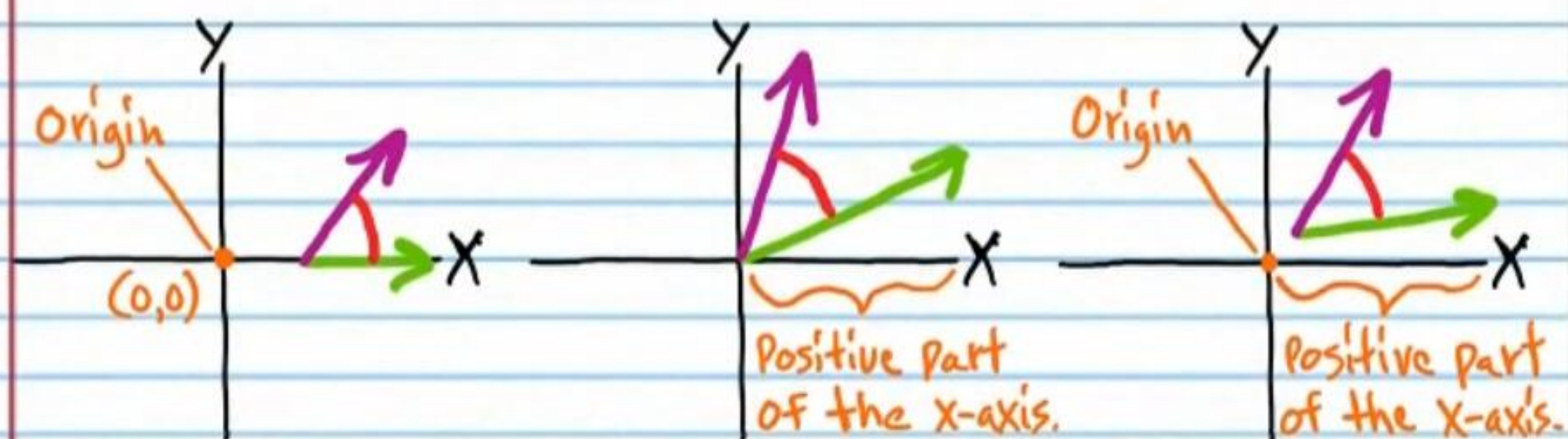
Angle Measure: The number of degrees (or radians) indicating the extent of rotation.

Examples of Angles that Are Not in Standard Position

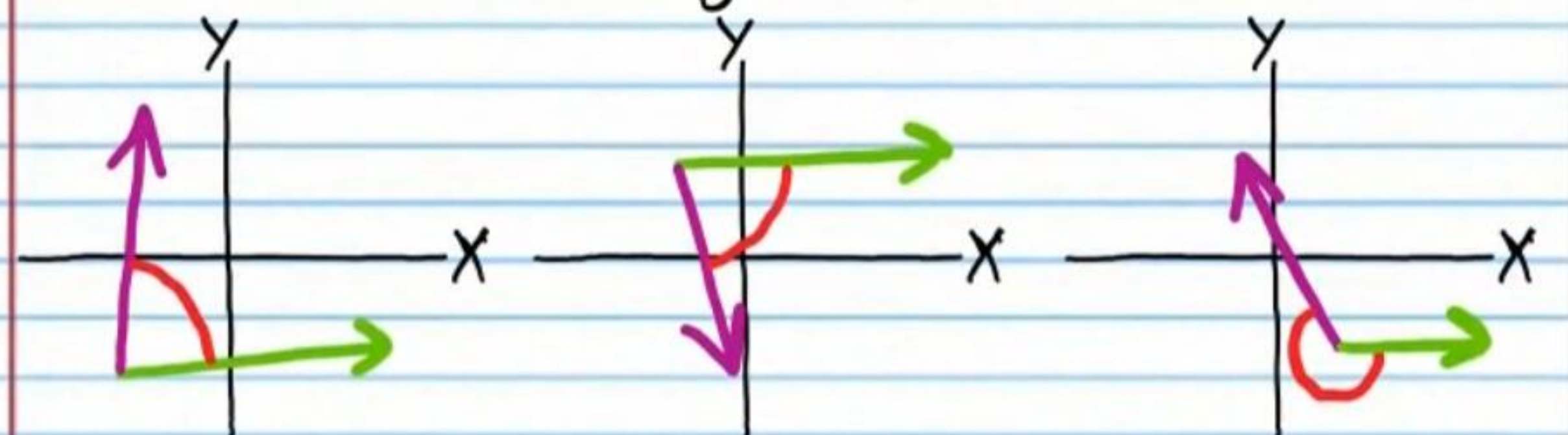
The vertex is not at the origin.

The initial side of the angle is not on the positive part of the x -axis.

The vertex is not at the origin and the initial side is not on the positive part of the x -axis.



Other Examples of Angles Not in Standard Position

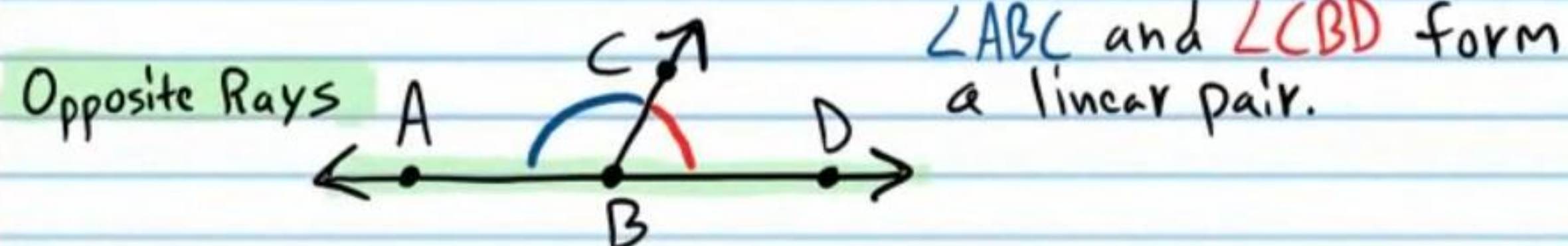


Practice

Draw angles with the following measures in standard position. Your drawings do not have to be exact, so a protractor is not necessary.

- ① 45° ② 180° ③ 340°
 - ④ 100° ⑤ 270° ⑥ 14°
-

Linear Pair: Two adjacent angles whose noncommon sides are opposite rays.



Linear Pair Postulate: If two angles form a linear pair, then they are supplementary.

Supplementary Angles: Two angles whose measures add up to 180° .

Practice

Find x .

• ⑦

$$x + 145 = 180$$

$$\boxed{x = 35^\circ}$$

⑧

$$x + 100 = 180$$

$$\boxed{x = 80^\circ}$$

• ⑨

$$x + 130 = 180$$

$$\boxed{x = 50^\circ}$$

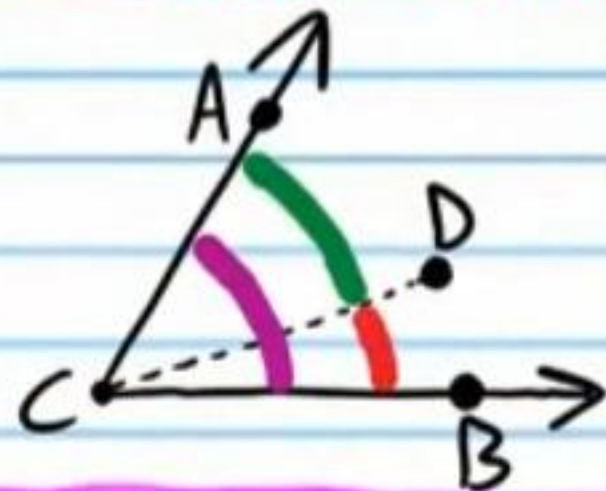
⑩

$$x + 110 = 180$$

$$\boxed{x = 70^\circ}$$

Angle Addition Postulate

If D is in the interior of $\angle ACB$, then
 $m\angle ACB = m\angle ACD + m\angle DCB$.



⑪

$$180 + X = 215$$

$$X = 35^\circ$$

⑫

$$305 + X = 360$$

$$X = 55^\circ$$

⑬

$$180 + X = 250$$

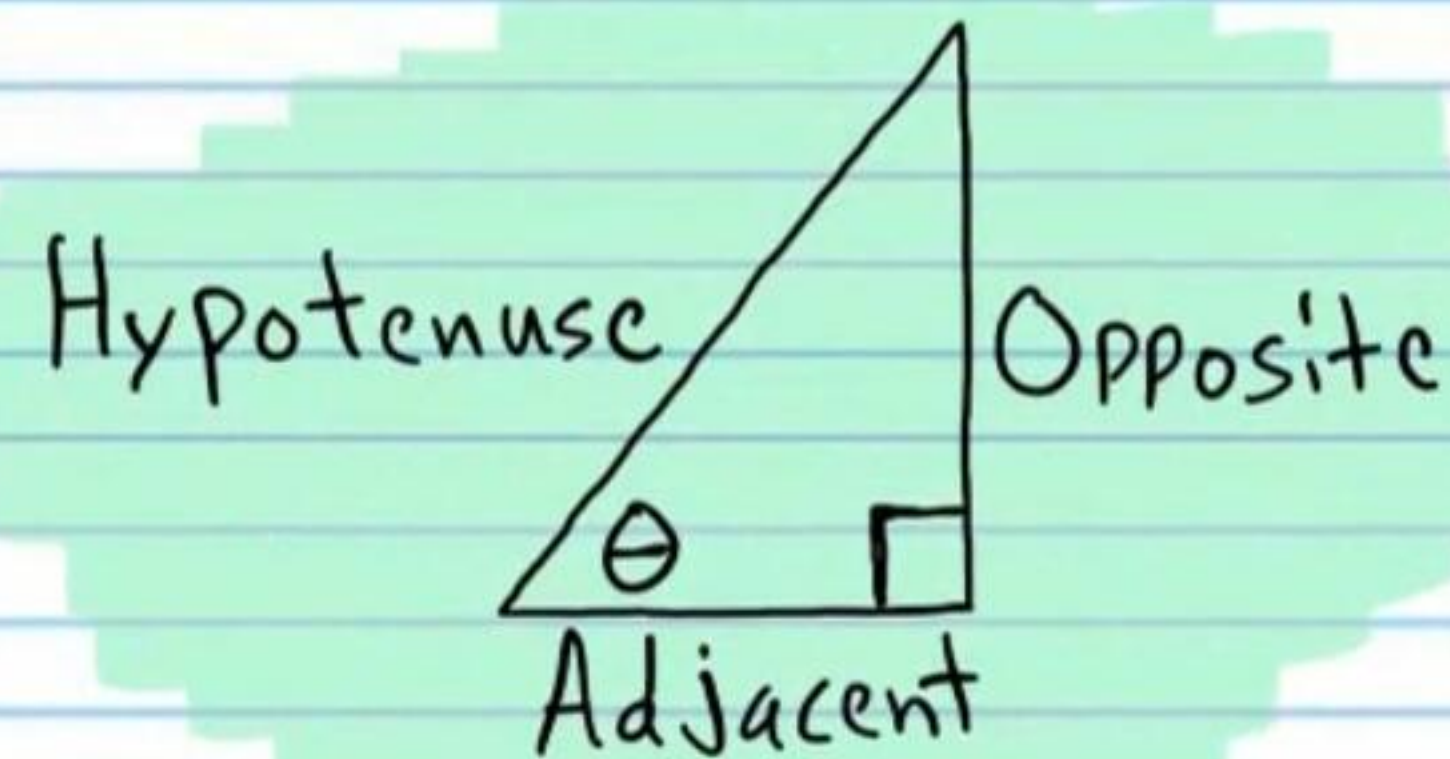
$$X = 70^\circ$$

⑭

$$360 = 335 + X$$

$$25^\circ = X$$

The Six Basic Trigonometric Functions



$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

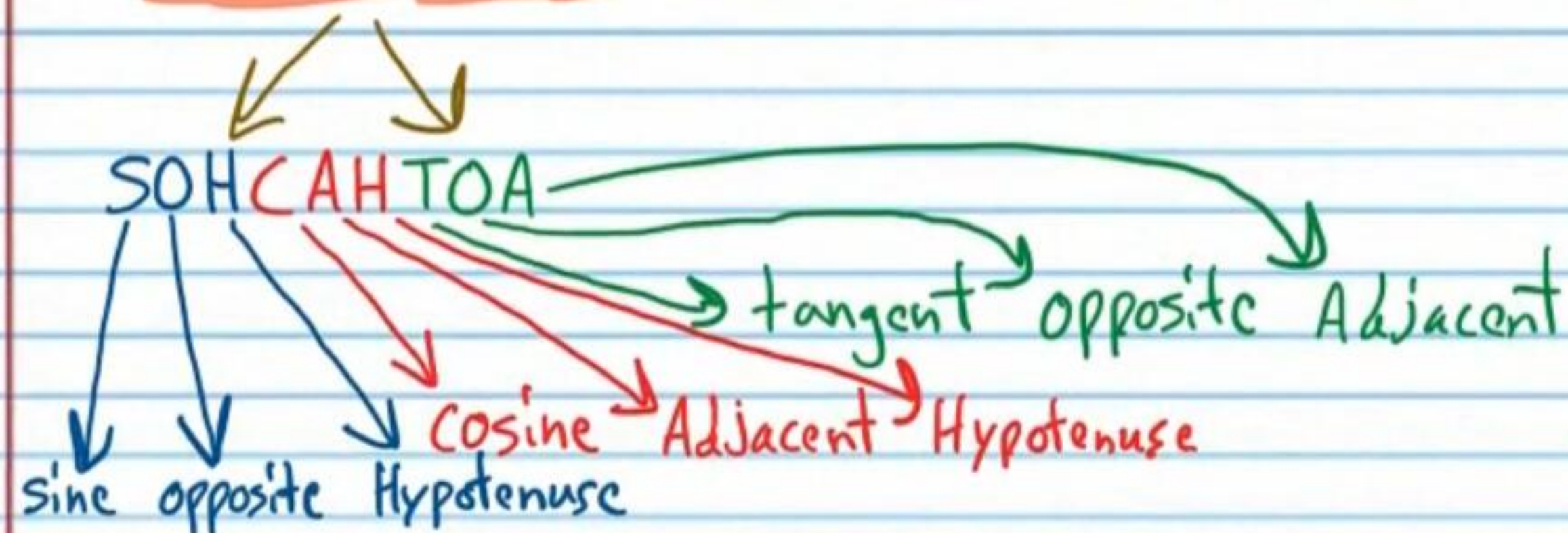
$$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

$$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$$

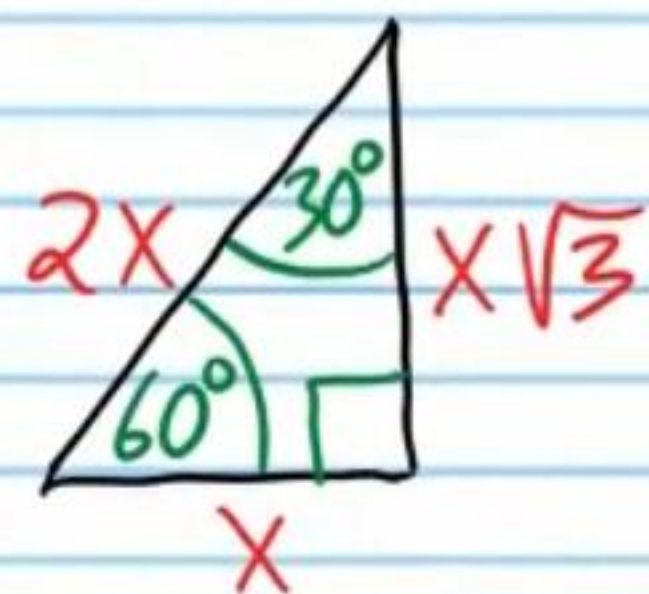
$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

$$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$$

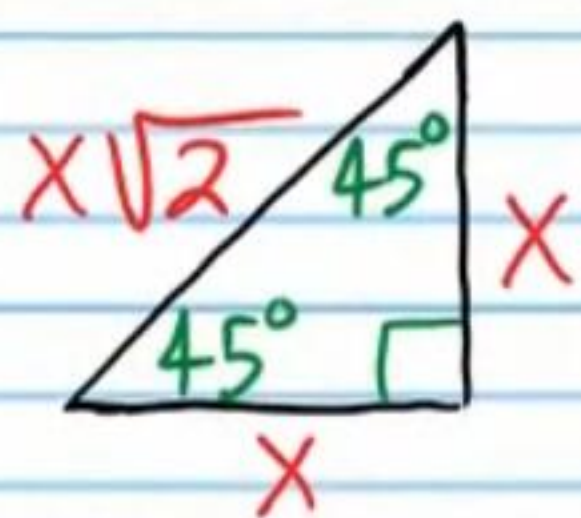


Special Right Triangles

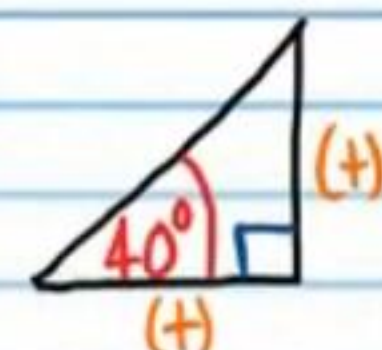
30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is X , then the length of the hypotenuse is $2X$ and the length of the side opposite the 60° angle is $X\sqrt{3}$.



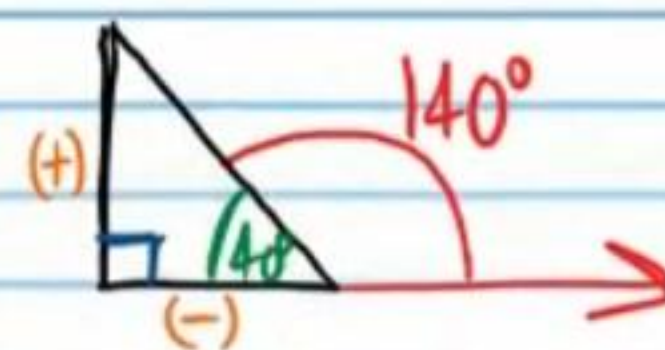
45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are X , then the length of the hypotenuse is $X\sqrt{2}$.



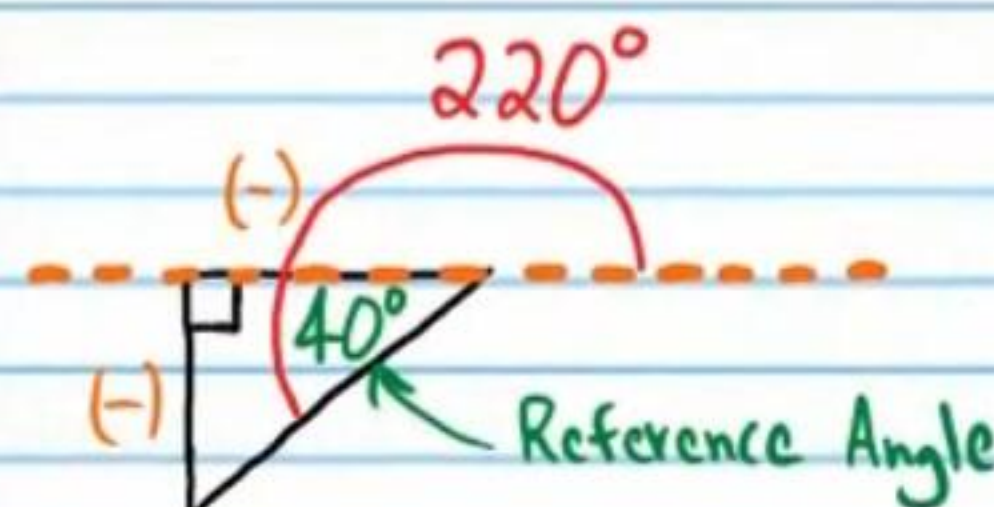
Trigonometric Expressions with Angle Measures Greater than 90°



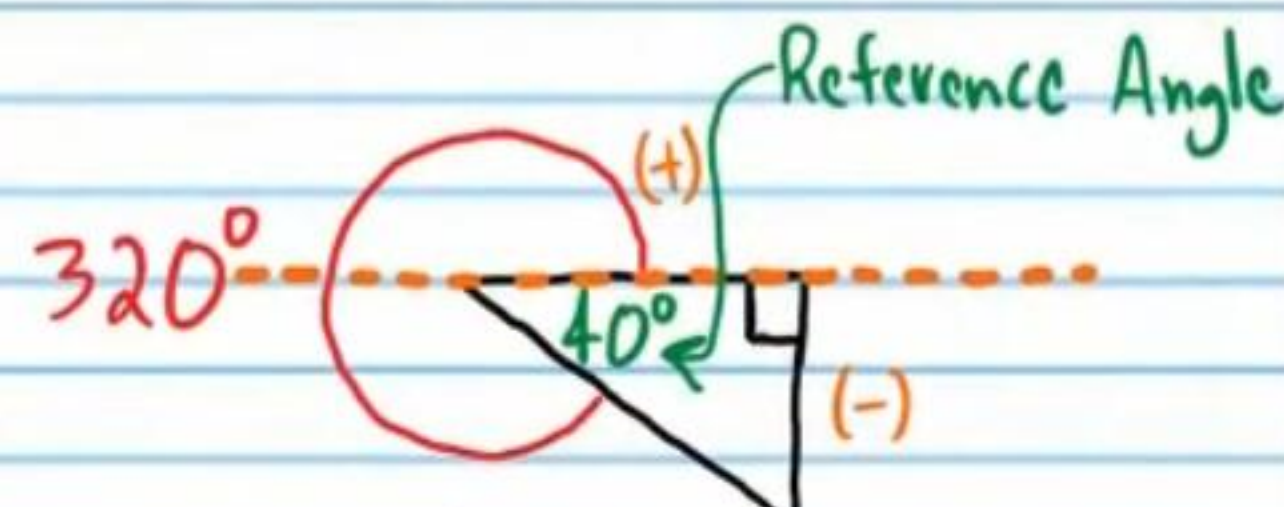
$$\sin 40^\circ \approx 0.6428$$



$$\sin 140^\circ \approx 0.6428$$

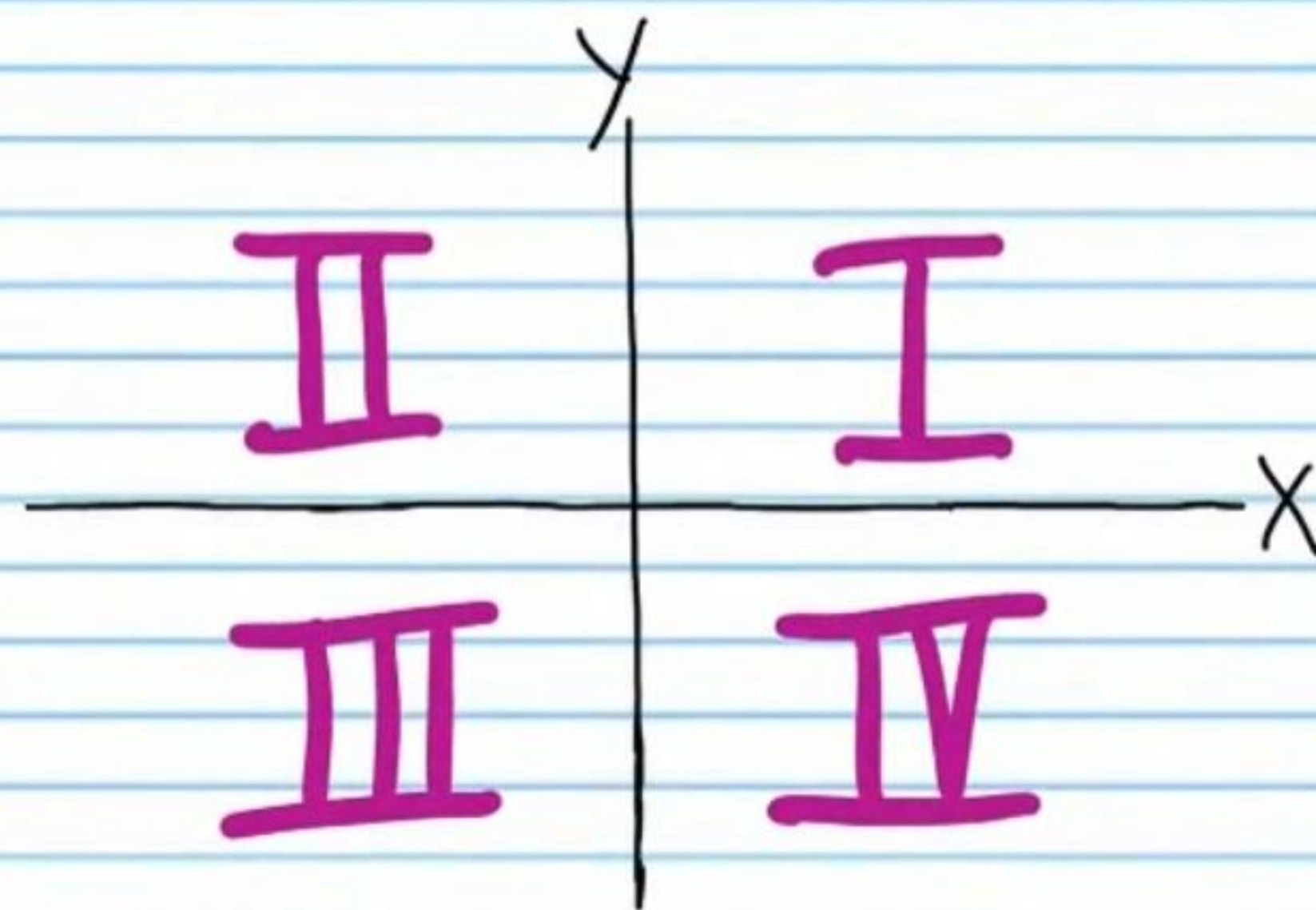


$$\sin 220^\circ \approx -0.6428$$



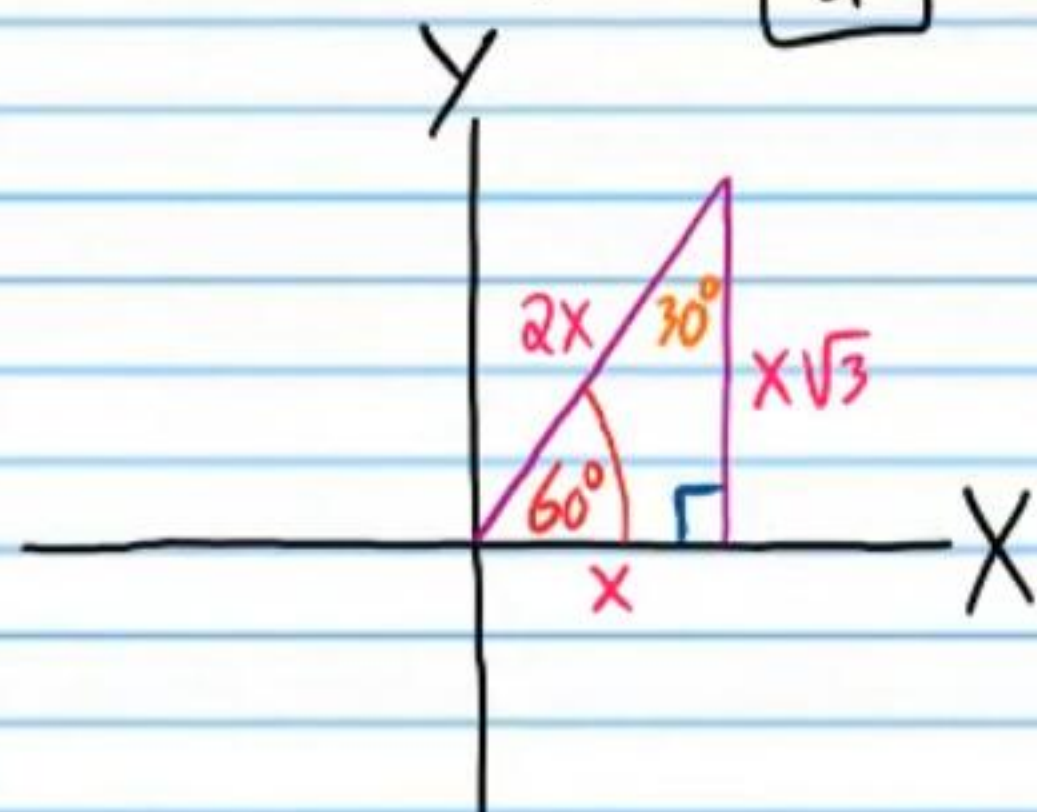
$$\sin 320^\circ \approx -0.6428$$

Quadrants of the XY Coordinate System

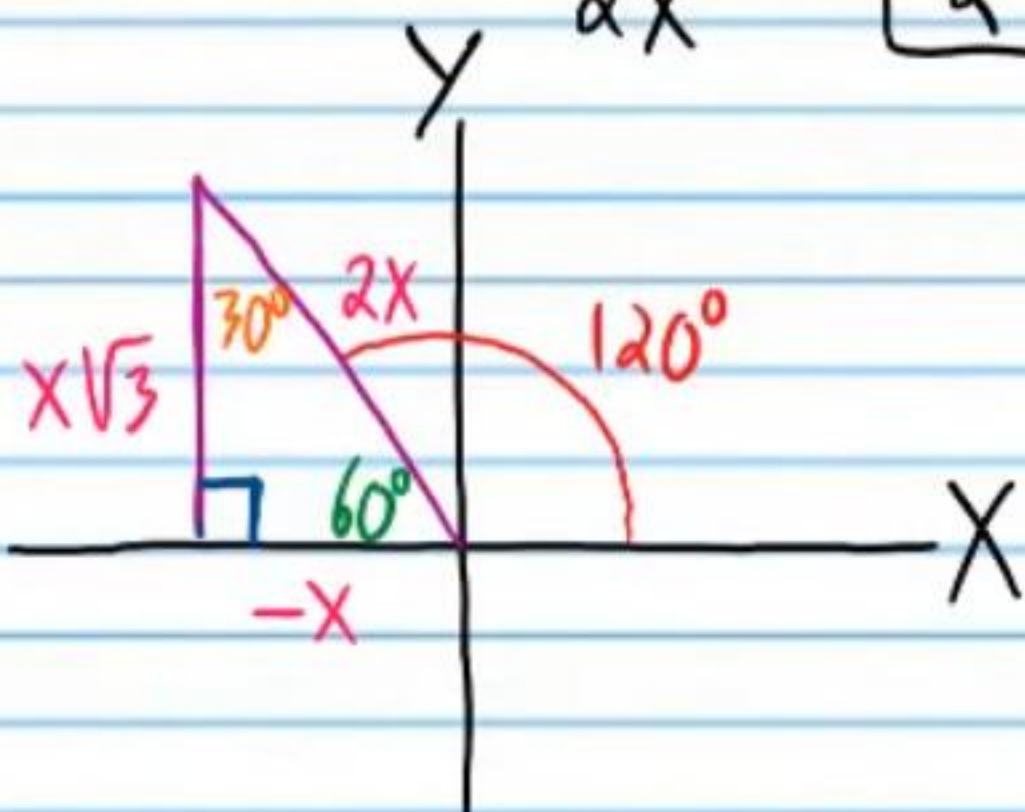


Example

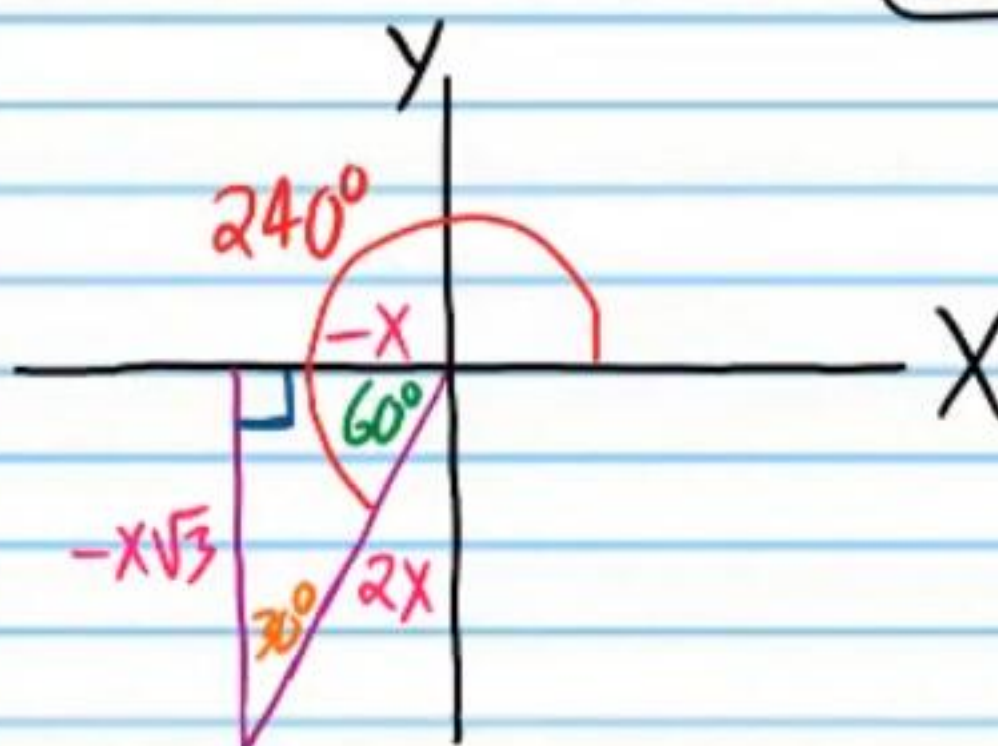
$$\sin 60^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$$



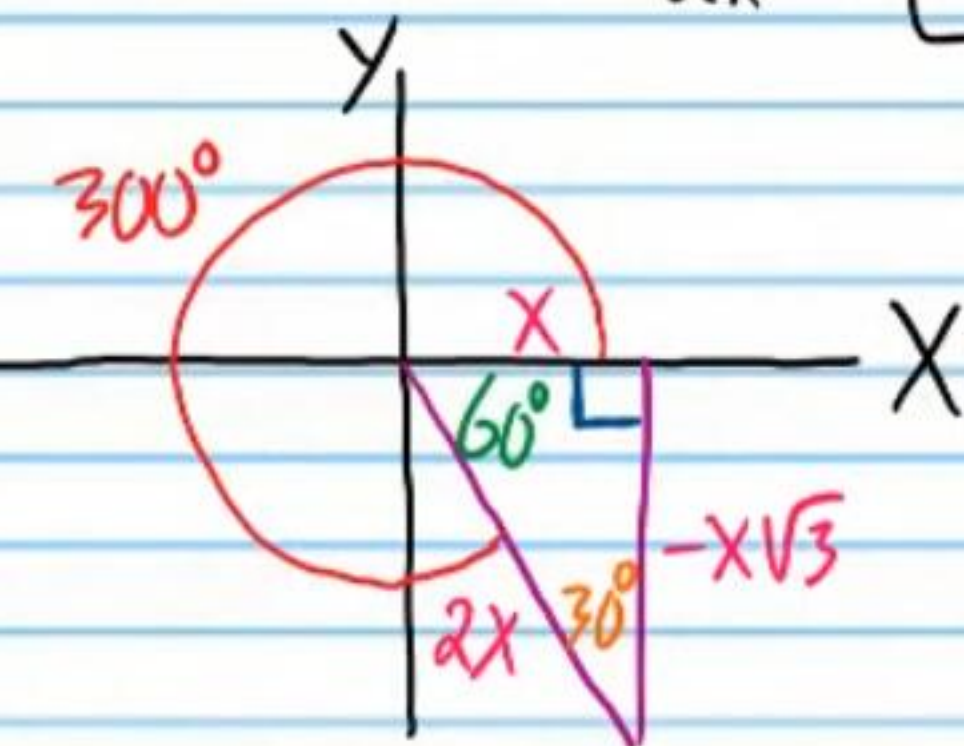
$$\sin 120^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$$



$$\sin 240^\circ = \frac{-x\sqrt{3}}{2x} = \boxed{-\frac{\sqrt{3}}{2}}$$



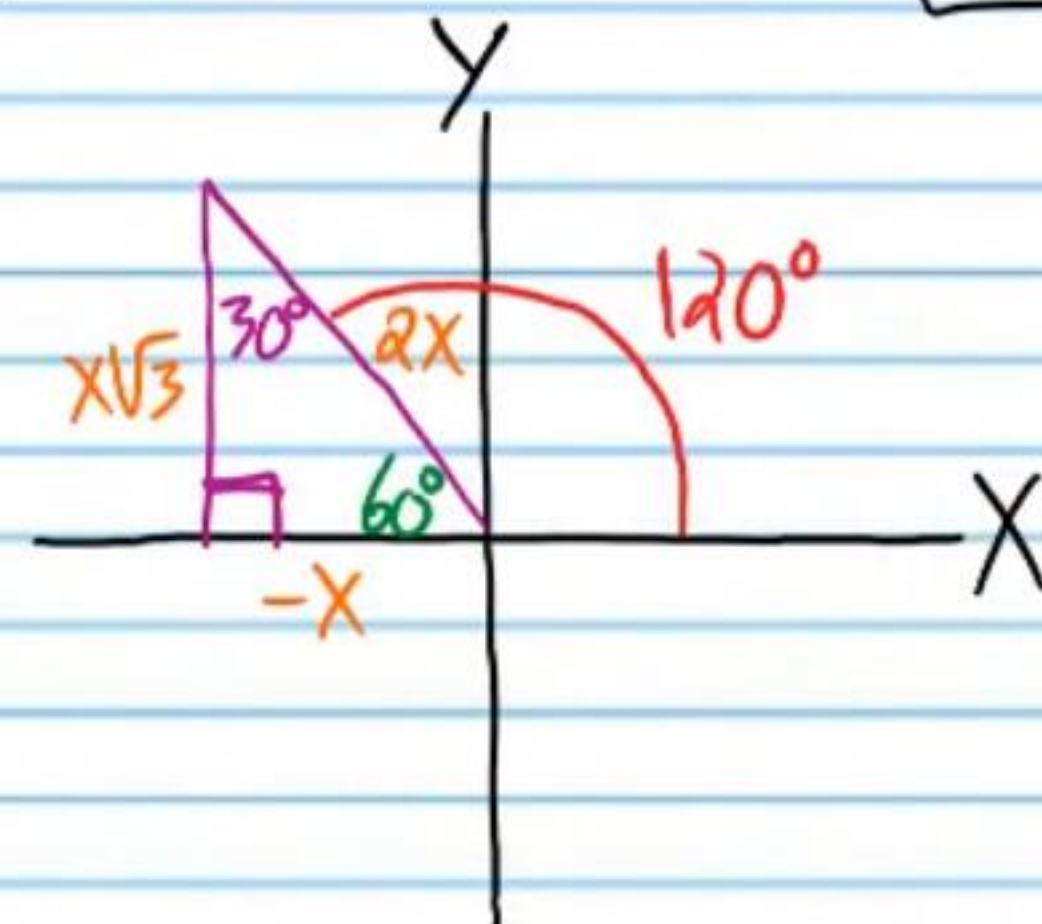
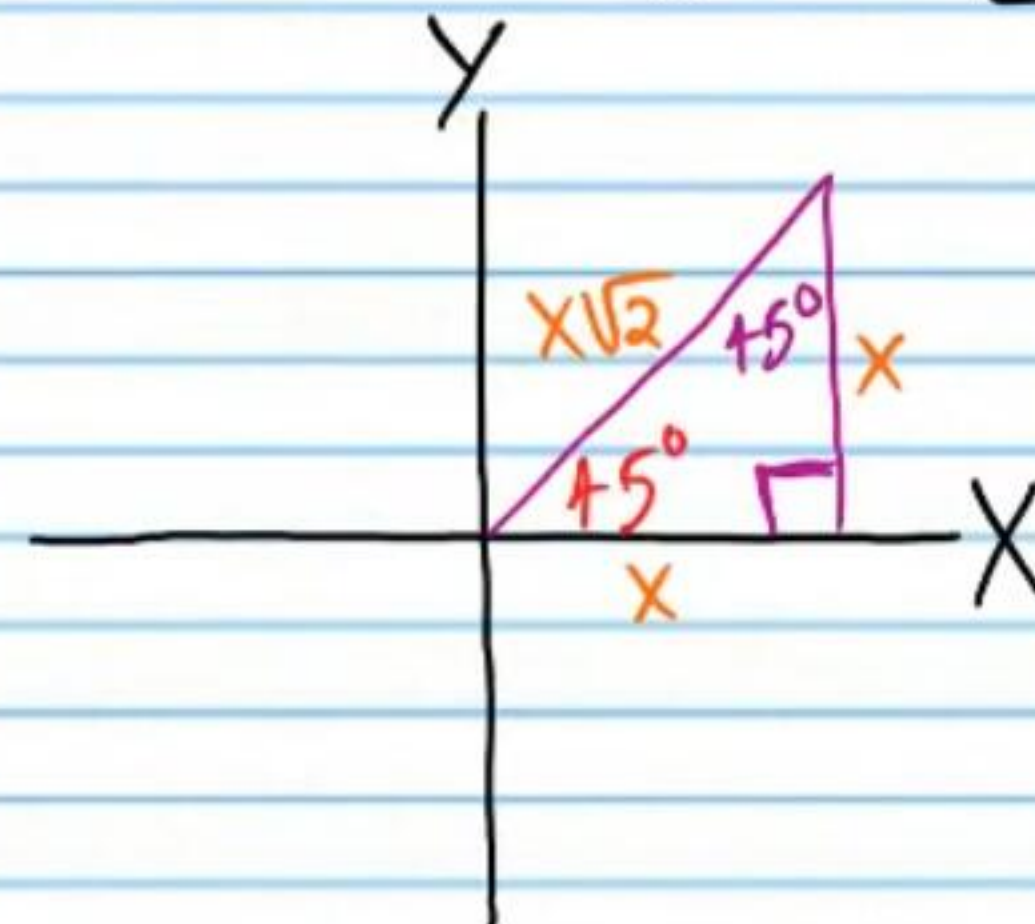
$$\sin 300^\circ = \frac{-x\sqrt{3}}{2x} = \boxed{-\frac{\sqrt{3}}{2}}$$



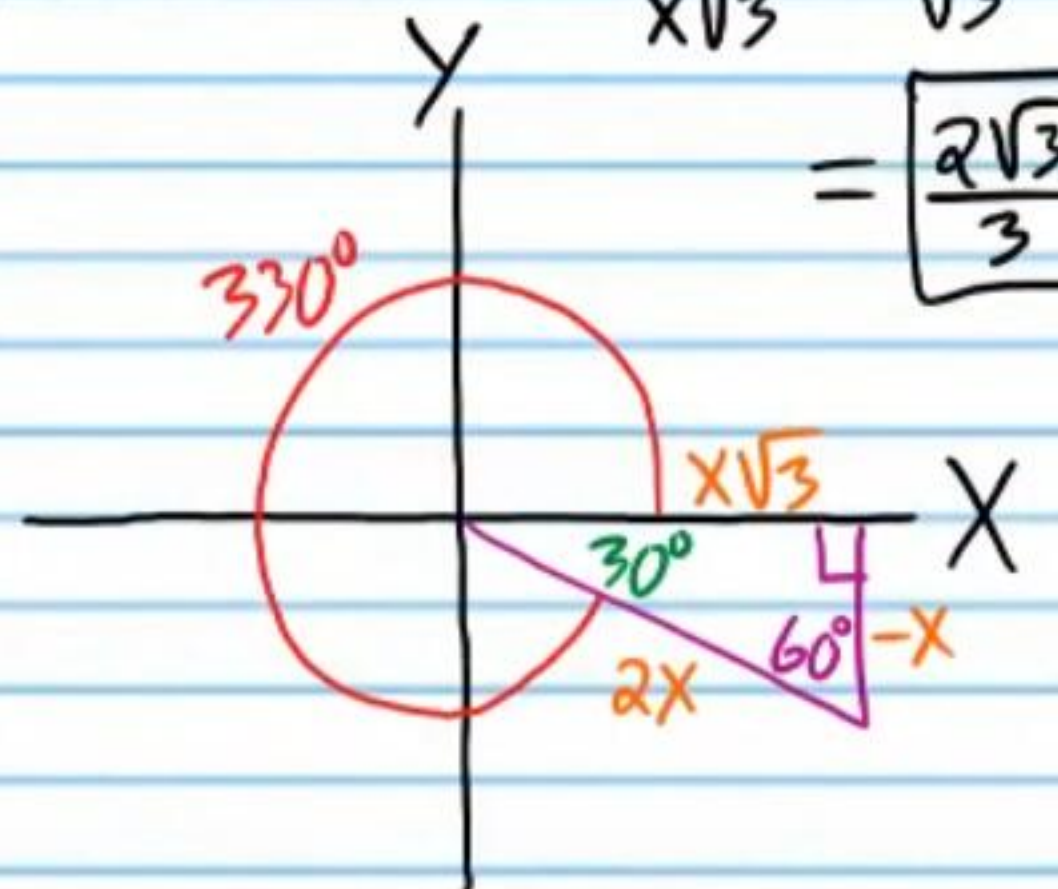
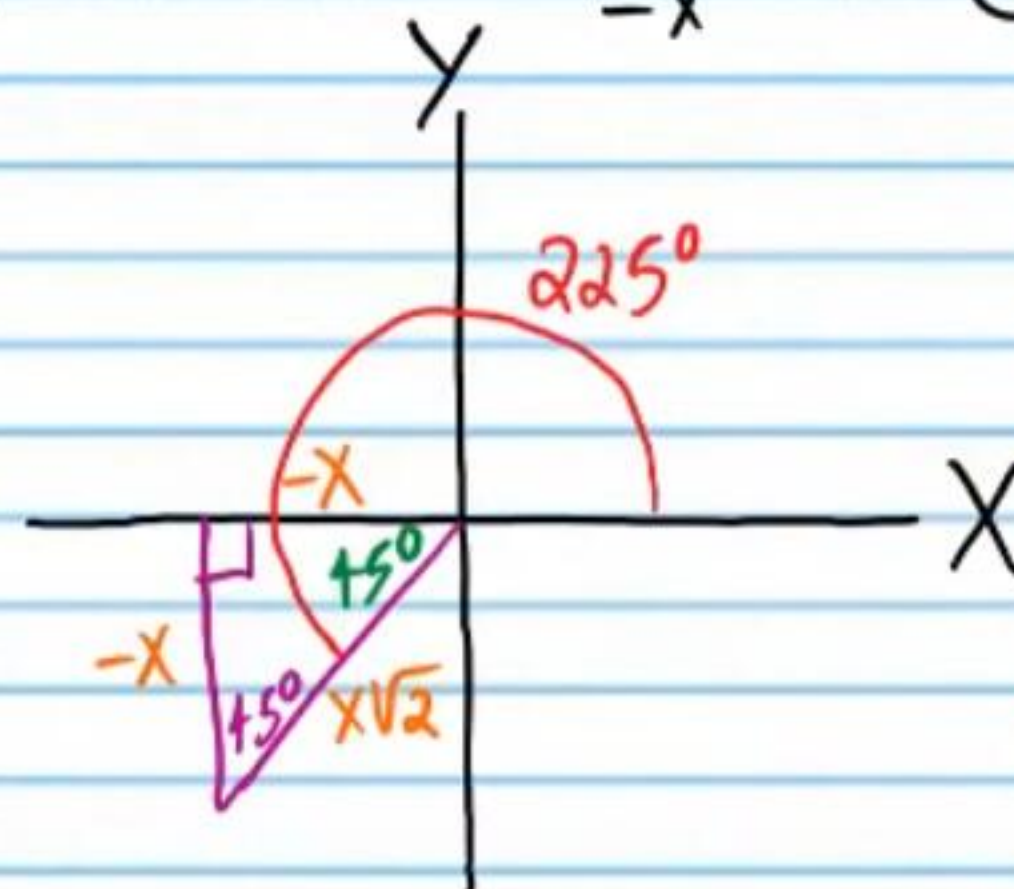
The Reference Angle of An Angle in Standard Position: The acute angle between the x-axis and the terminal side.

Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help find the value of the trigonometric expression. Then, find the value of the trigonometric expression *without a calculator*.

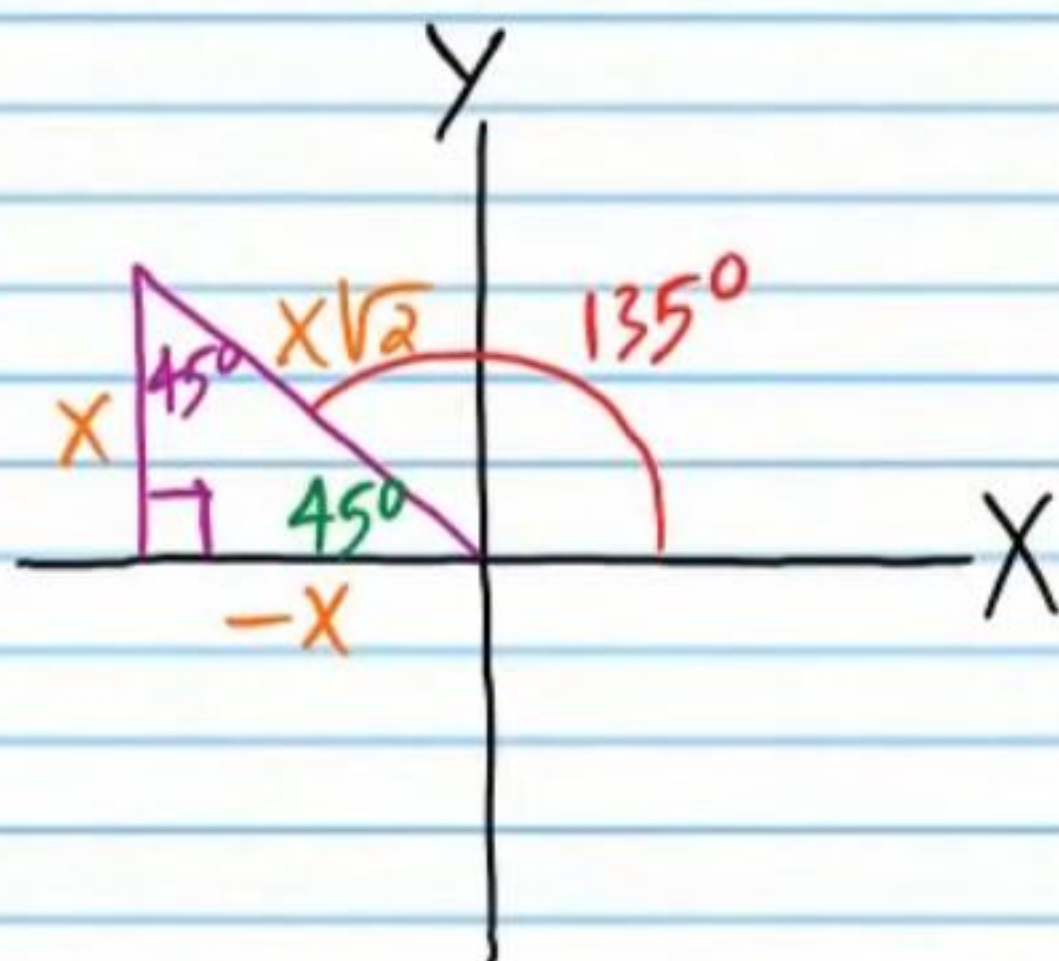
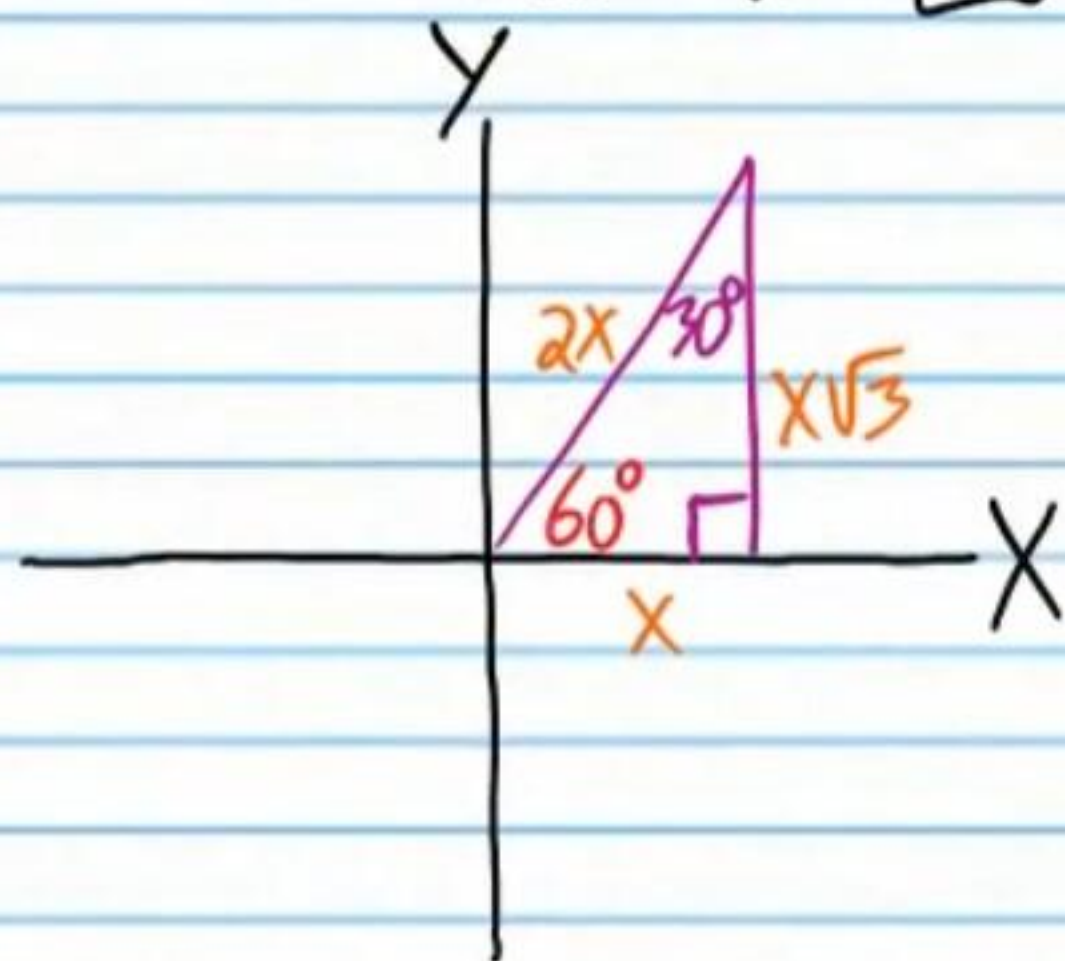
• ⑮ $\sin 45^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$ ⑯ $\cos 120^\circ = \frac{-x}{2x} = \boxed{-\frac{1}{2}}$



⑰ $\tan 225^\circ = \frac{-x}{-x} = \boxed{1}$ ⑱ $\sec 330^\circ = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$

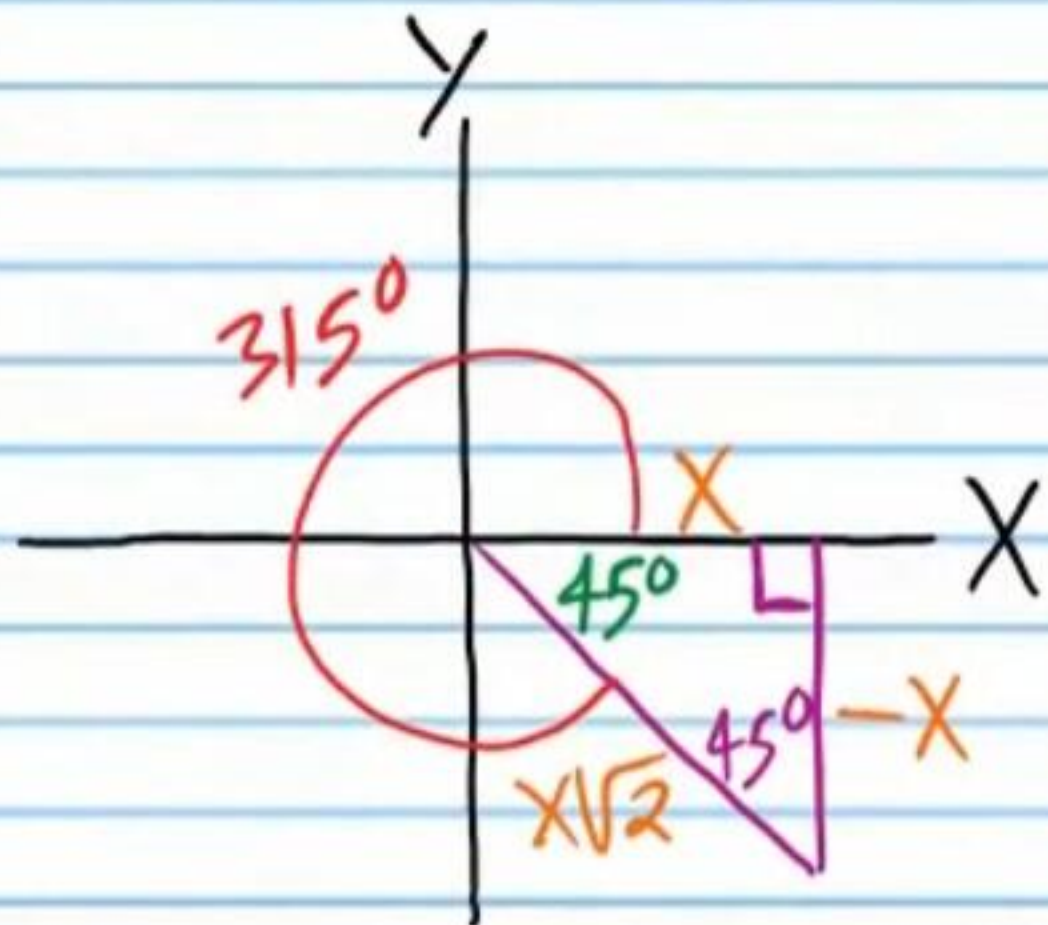
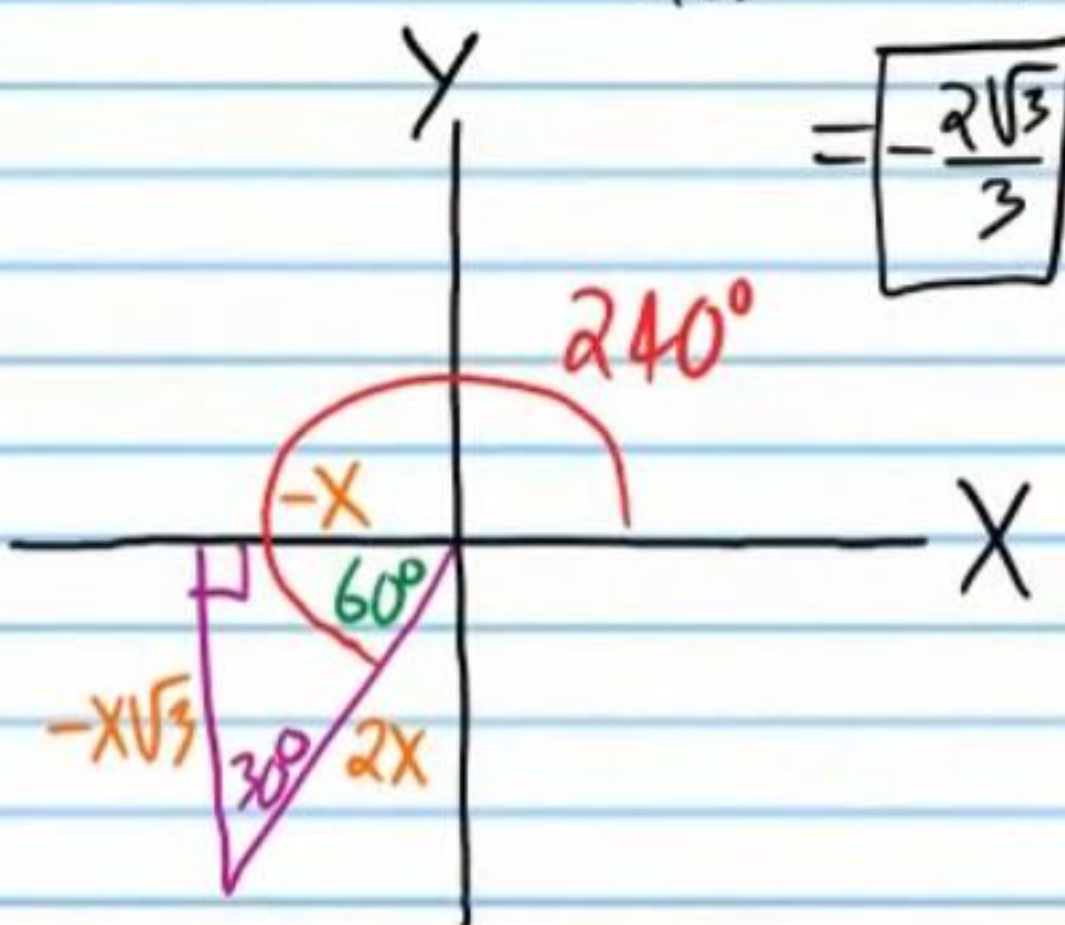


• (19) $\cot 60^\circ = \frac{x}{x\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$ (20) $\sin 135^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$



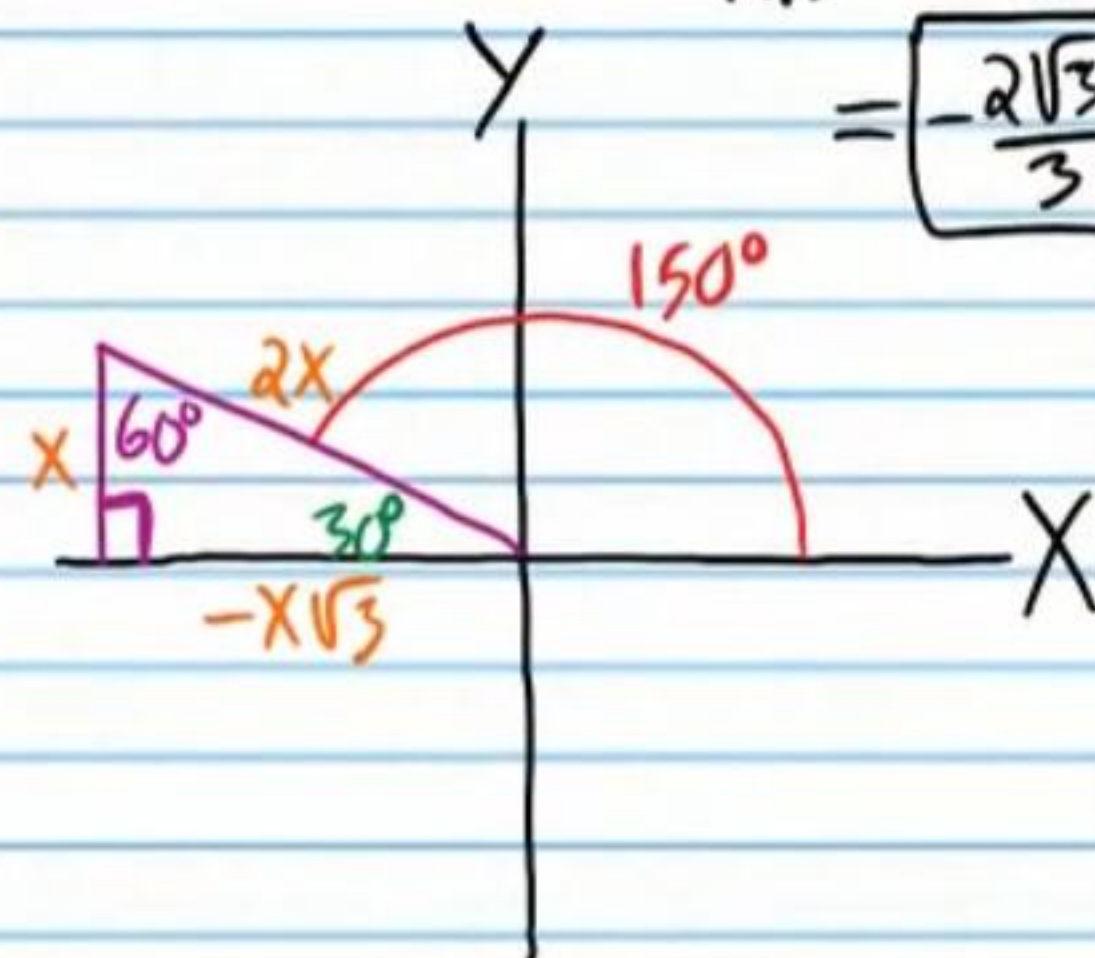
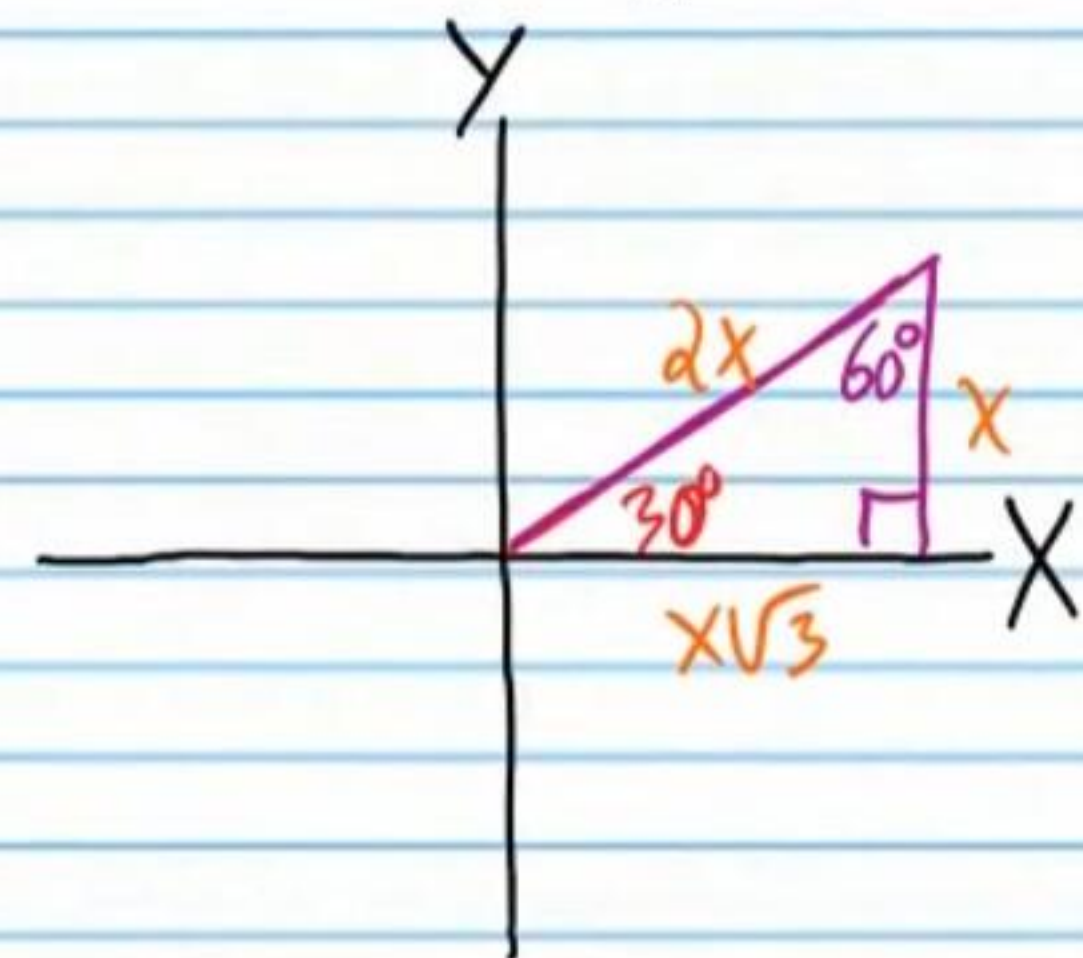
(21) $\csc 240^\circ = \frac{2x}{-x\sqrt{3}} = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$

(22) $\tan 315^\circ = \frac{-x}{x} = \boxed{-1}$

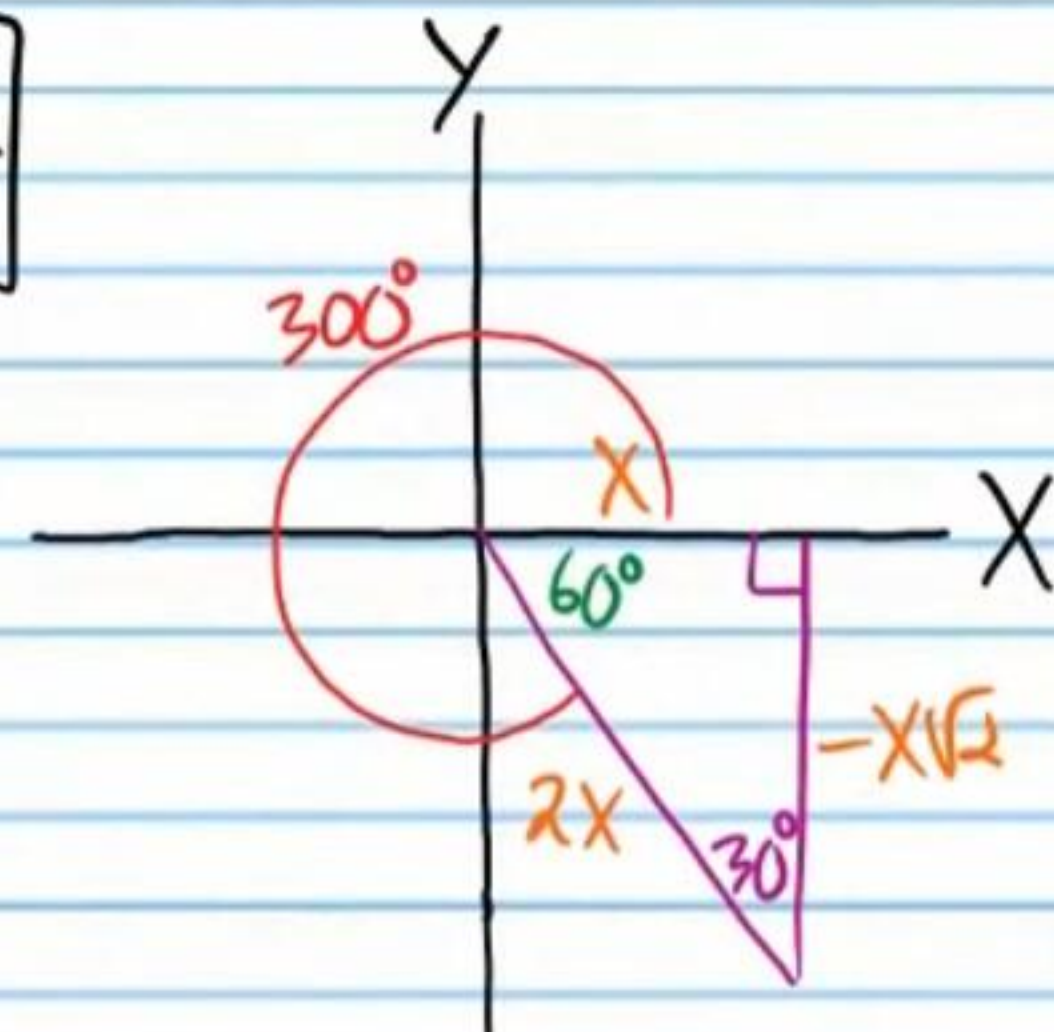
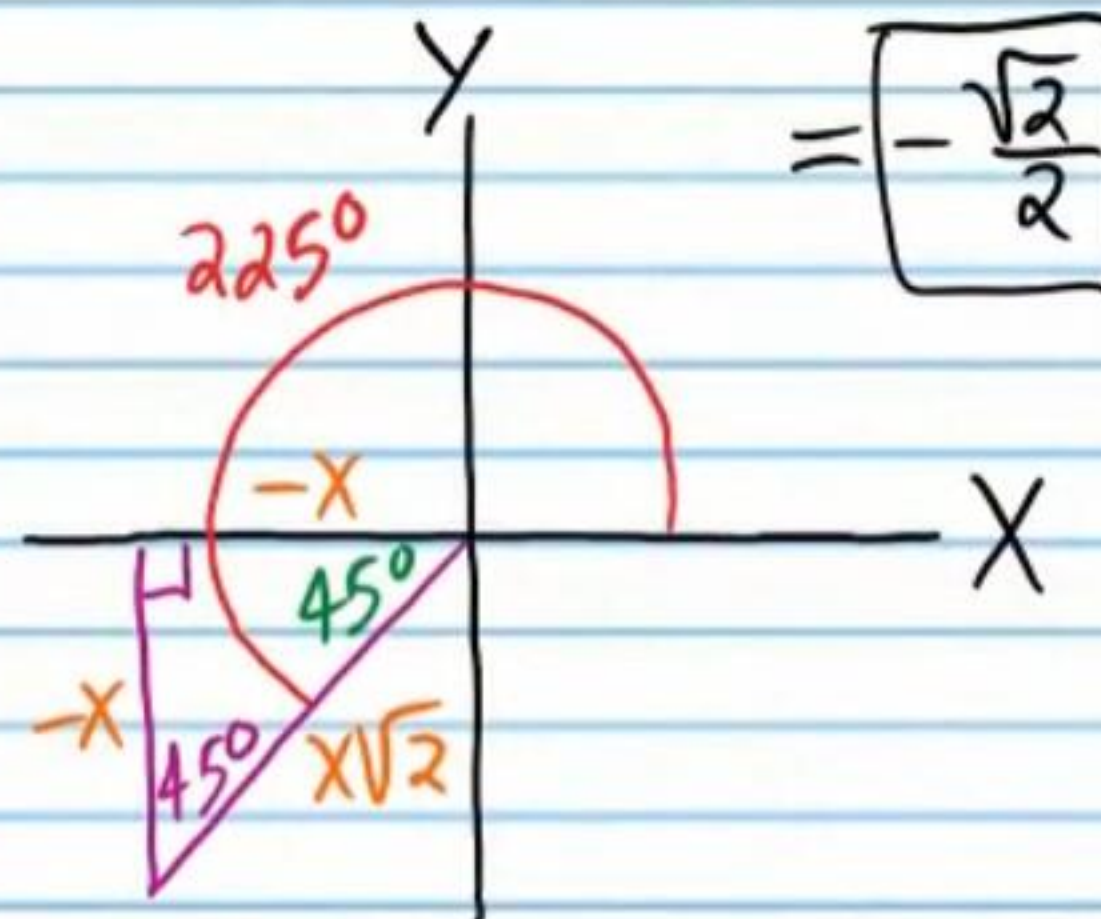


(23) $\csc 30^\circ = \frac{2x}{x} = \boxed{2}$

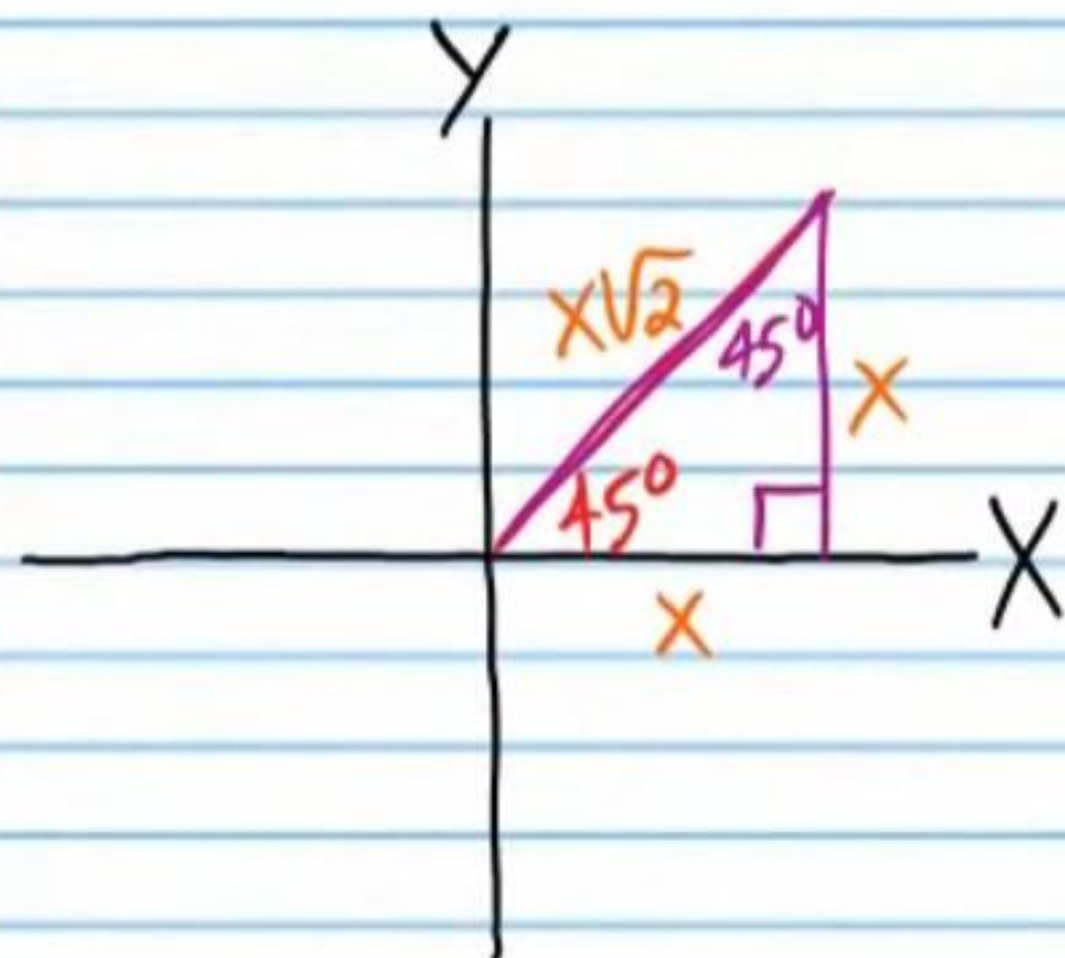
(24) $\sec 150^\circ = \frac{2x}{-x\sqrt{3}} = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$



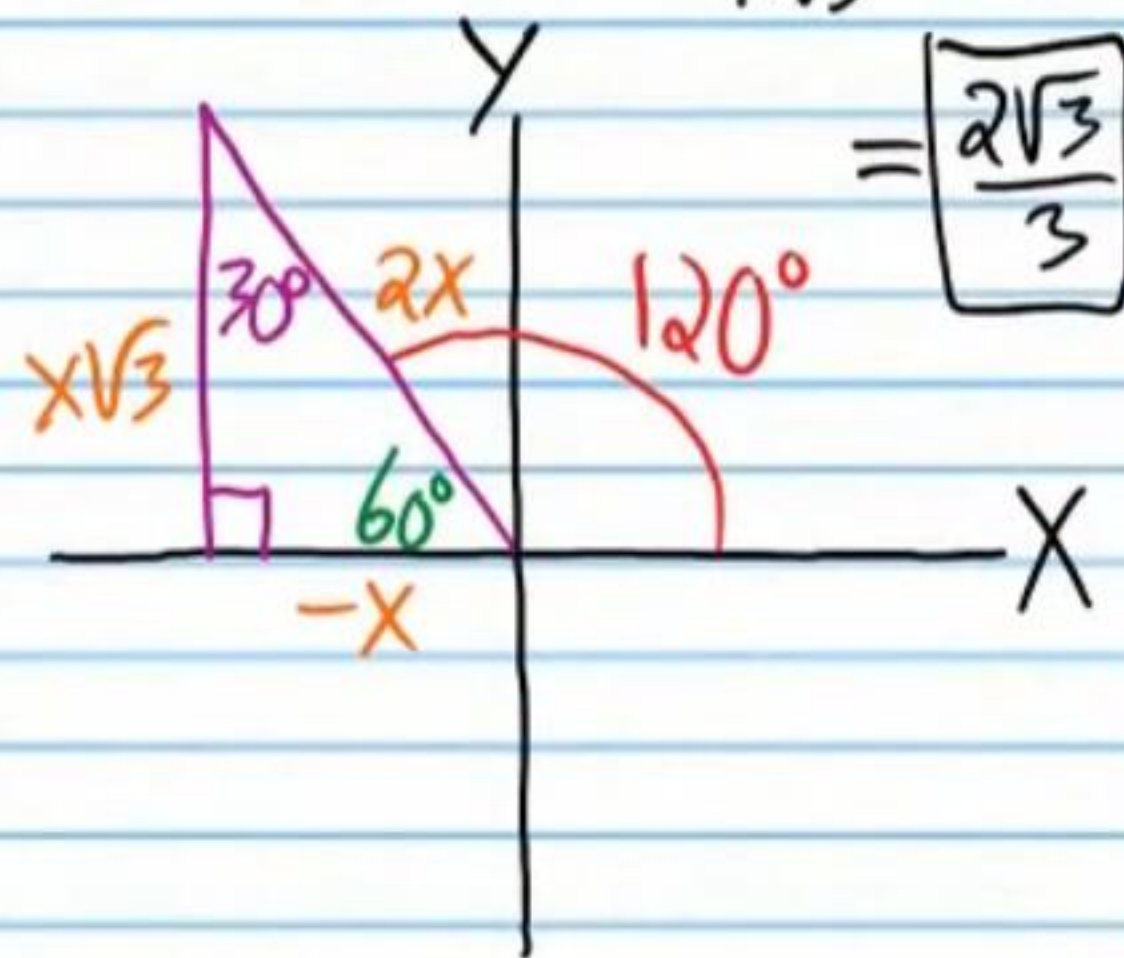
(25) $\sin 225^\circ = \frac{-x}{x\sqrt{2}} = -\frac{1}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}$ (26) $\cos 300^\circ = \frac{x}{2x} = \boxed{\frac{1}{2}}$



$$\textcircled{27} \tan 45^\circ = \frac{x}{x} = \boxed{1}$$

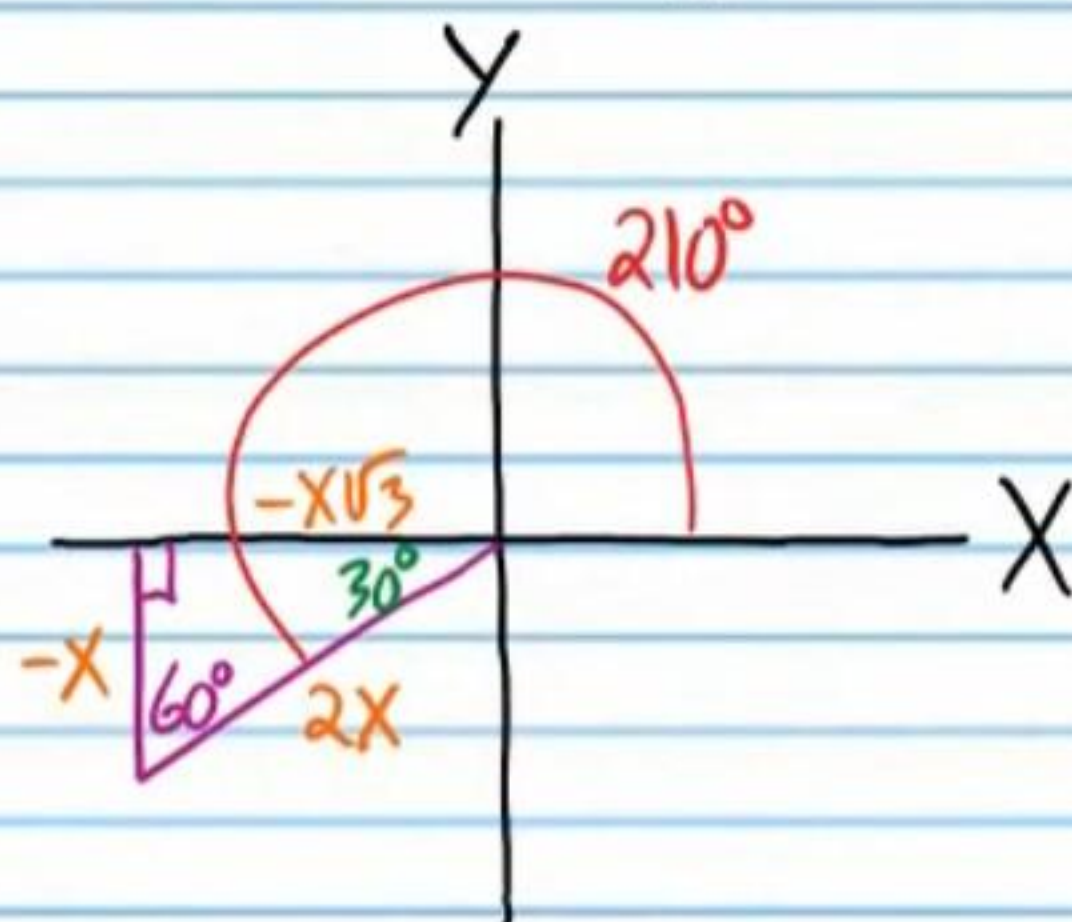


$$\textcircled{28} \csc 120^\circ = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}}$$

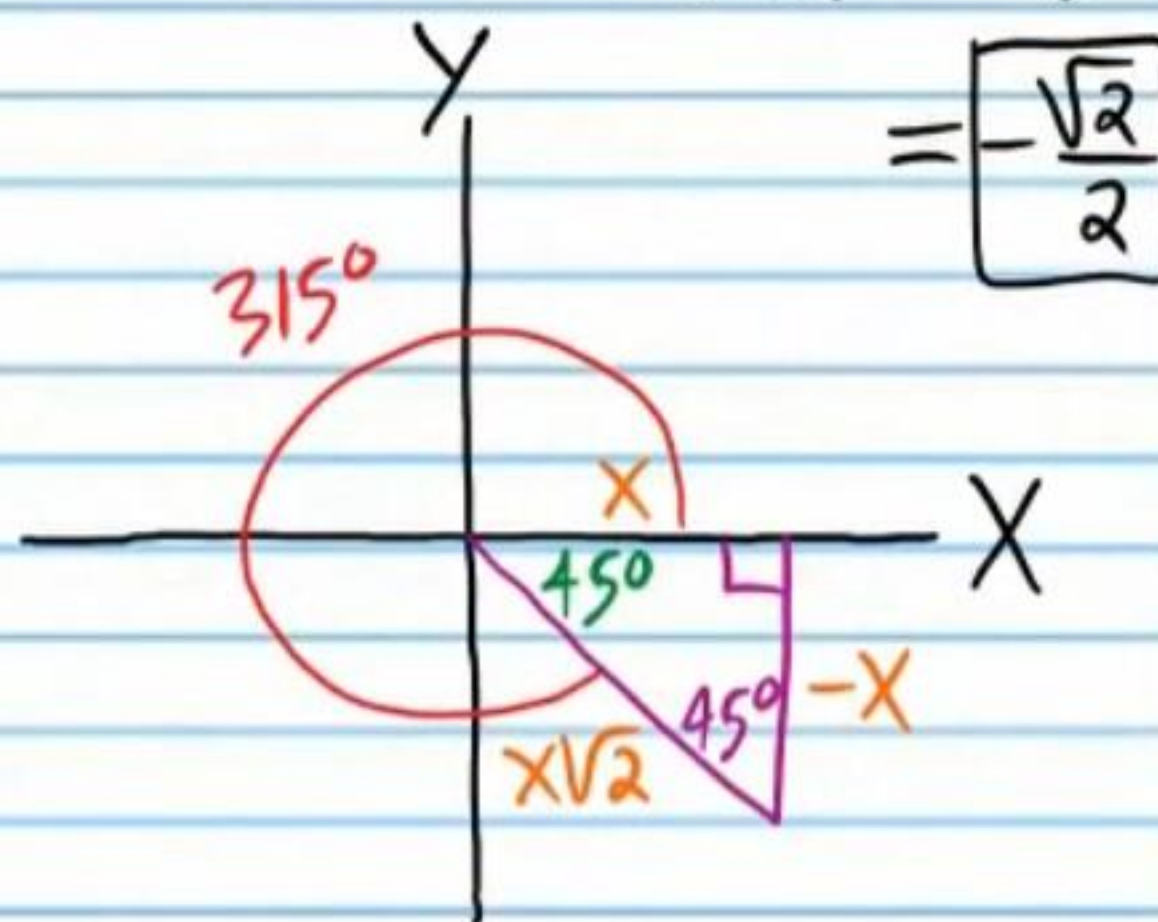


$$= \boxed{\frac{2\sqrt{3}}{3}}$$

$$\textcircled{29} \cot 210^\circ = \frac{-x\sqrt{3}}{-x} = \boxed{\sqrt{3}}$$



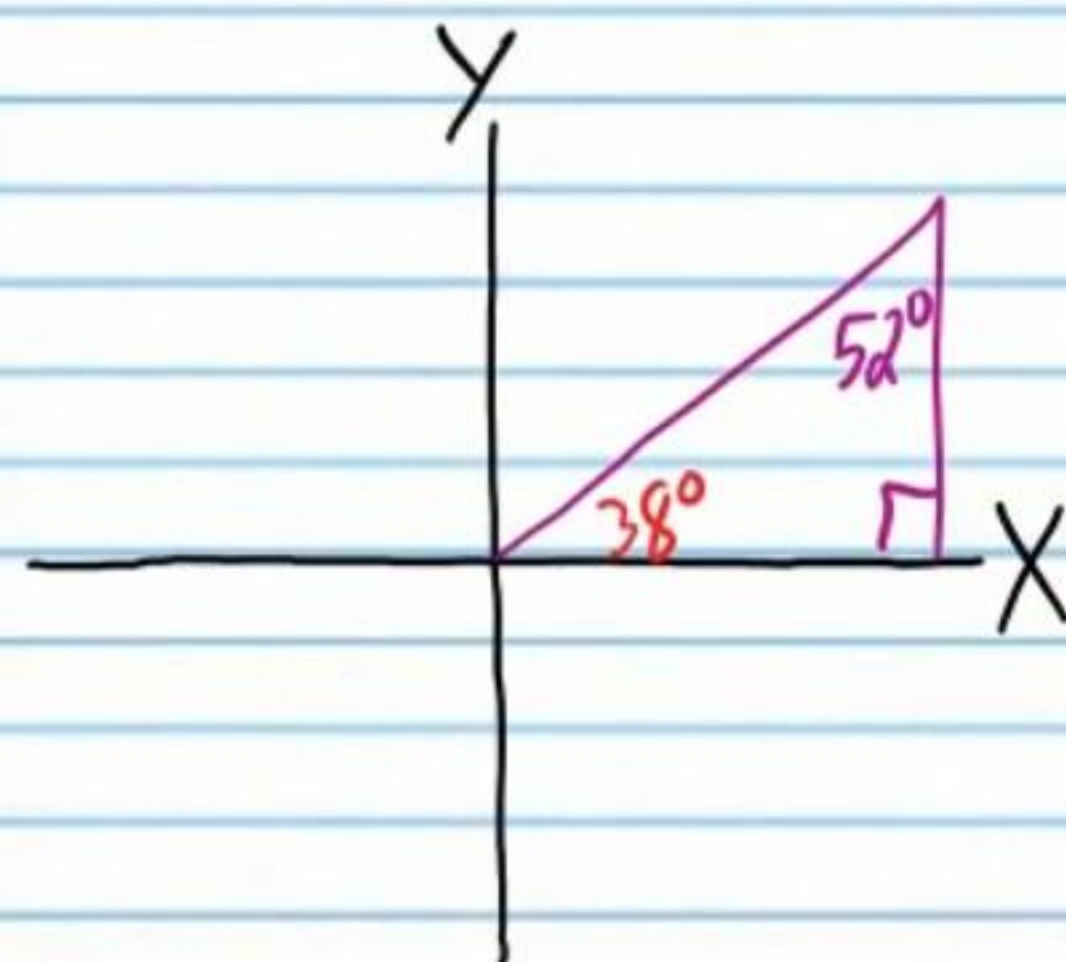
$$\textcircled{30} \sin 315^\circ = \frac{-x}{x\sqrt{2}} = -\frac{1}{\sqrt{2}}$$



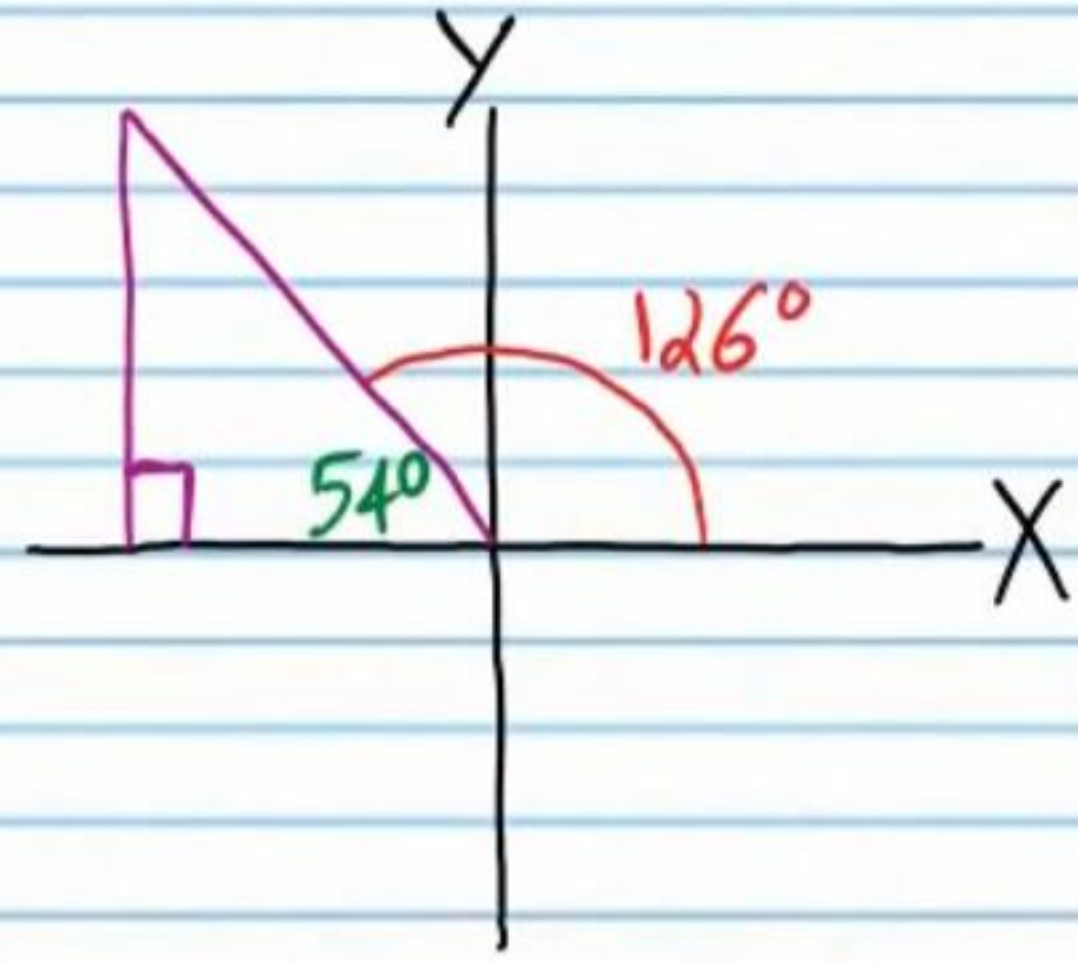
$$= \boxed{-\frac{\sqrt{2}}{2}}$$

Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help understand the value of the trigonometric expression. Then, use a calculator to determine the values of each trigonometric expression. Round to four decimal places. Explain why some values are positive and some are negative.

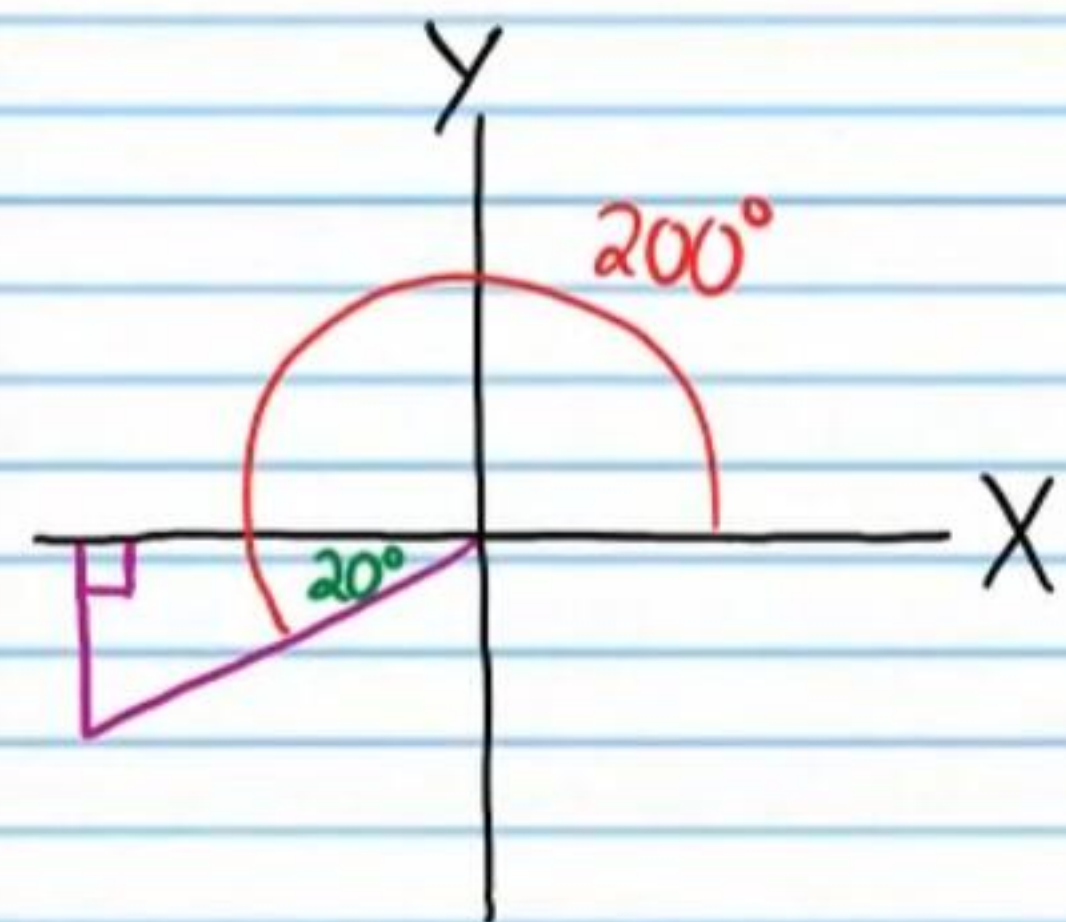
$$\bullet \textcircled{31} \cos 38^\circ \approx \boxed{0.7880}$$



$$\textcircled{32} \tan 126^\circ \approx \boxed{-1.3764}$$



$$\bullet \textcircled{33} \sin 200^\circ \approx \boxed{-0.3420}$$



$$\textcircled{34} \cos 325^\circ \approx \boxed{0.8192}$$

